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Studierichting: Informatica
Oefeningenreeks 3C nr. 17.7

$$
\begin{aligned}
&\lim_{x\to 0} \frac{x\sin^2 x}{x\cos x - \sin x} = \frac{0}{0} \\
&\overset{H}{=} \lim_{x\to 0} \frac{\sin^2 x + x(2\sin x\cos x)}{\cos x + x(-\sin x) - \cos x} \\
&= \lim_{x\to 0} \frac{\sin^2 x + x\sin(2x)}{-x\sin x} = \frac{0}{0} \\
&= \lim_{x\to 0} \frac{\sin^2 x + x(2\sin x\cos x)}{-x\sin x} = \lim_{x\to 0} \frac{\sin x + 2x\cos x}{-x} \\
&\overset{H}{=} \lim_{x\to 0} \frac{3\cos x - 2x\sin x}{-1} = -3
\end{aligned}
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References